

Structural $\dim = 2$ for $(3, 3, 1^q)$: outer-factor injectivity via adjacent disjointness

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Abstract

The same-row-dies paper of 13 May night (2026-05-13-same-row-dies-331.tex, SRD-331) proved that for $\lambda = (3, 3, 1^q)$, $q \geq 2$,

$$B_{\ell+1}^{(\lambda)} V_\lambda = (T_\ell+1)(T_{\ell+1}+1)(T_{n-2}+1) \Phi_* \left(B_{\ell(\mu_1)+1}^{(\mu_1)} V_{\mu_1} \right)$$

with $\mu_1 = (3, 2, 1^{q-1})$ on $n-2 = q+4$ letters. Combined with the May-12 evening theorem $\dim B^{(\mu_1)} = 2$ ($q \geq 4$), this gives the upper bound $\dim B^{(\lambda)} \leq 2$ structurally at $q \geq 4$. The matching structural lower bound $\dim B^{(\lambda)} \geq 2$ was open (SRD-331 Remark 4.6); only computational verification at $q \in \{2, 3, 4, 5\}$ was available.

Theorem. For $\lambda = (3, 3, 1^q)$ with $q \geq 4$,

$$\dim B_{\ell+1}^{(\lambda)} V_\lambda = 2.$$

Proof technique. The new ingredient is a clean injectivity argument for the outer factors. Each factor (T_j+1) maps $E_{j+1}^+|_{V_\lambda}$ *injectively* into $E_j^+|_{V_\lambda}$, because (i) its image lies in E_j^+ by the quadratic relation $(T_j-q)(T_j+1) = 0$, and (ii) its kernel E_j^- intersects E_{j+1}^+ trivially by the May-19 abstract adjacent-disjointness lemma (in its symmetrically swapped form). Chaining the three factors gives an injective composition $E_{n-1}^+ \rightarrow E_{n-2}^+ \rightarrow E_{n-3}^+ \rightarrow E_{n-4}^+$. Since $\Phi_*(B^{(\mu_1)}) \subseteq E_{n-1}^+$ has $\dim 2$, its image has $\dim 2$, giving the lower bound.

What this generalises. The proof works for any $r = 1$ rank-zero shape λ where (a) the SRD reduction formula applies (same-row-dies holds and the corner-pair 2-block is non-degenerate), and (b) $\dim B^{(\mu_1)}$ is known. The upper-and-lower bound then collapses to the equality $\dim B^{(\lambda)} = \dim B^{(\mu_1)}$. Specialising to $\lambda = (3, 3, 1^q)$ with $\mu_1 = (3, 2, 1^{q-1})$ and May-12 evening's $\dim = 2$ gives this paper's main theorem.

Computational verification. The injectivity chain $E_j^- \cap E_{j+1}^+ = 0$ verified for $j \in \{n-4, n-3, n-2\}$ and the outer-chain rank on E_{n-1}^+ equals $\dim E_{n-1}^+$ at $q \in \{4, 5, 6\}$.

1 Setup and statement

Notation follows the SRD-331 paper. $H_q(S_n)$ is the Iwahori–Hecke algebra over $\mathbb{Q}(q)$; V_λ is the seminormal cell module for $\lambda \vdash n$; $E_j^\pm$ are the q - and (-1) -eigenspaces of T_j on V_λ ;

$$R'_k := (T_{k-1}+1) \cdots (T_1+1), \quad B_j^{(\lambda)} V_\lambda := R'_j \cdots R'_n V_\lambda.$$

Throughout this paper, $\lambda := (3, 3, 1^q)$ with $q \geq 4$, $n := q+6$, $\ell := q+2$, $j_0 := \ell+1 = q+3 = n-3$. The corners of λ are $c_1 := (2, 3)$ (length-preserving) and $c_* := (q+2, 1)$ (column-1 bottom). Let $\mu_1 := \lambda \setminus \{c_1, c_*\} = (3, 2, 1^{q-1}) \vdash n-2$, which has $\ell(\mu_1) = q+1$ and $|\mu_1| = q+4 \geq 8$ when $q \geq 4$. The axial distance of the corner-pair 2-block is $d := c(c_*) - c(c_1) = (1 - (q+2)) - (3-2) = -(q+2)$, so $|d| \geq 6$ at $q \geq 4$ (in particular non-degenerate).

The branching iso Φ_* . The pair $\{c_1, c_*\}$ gives an $H_q(S_{n-2})$ -equivariant injection $\Phi_* : V_{\mu_1}|_{n-2} \hookrightarrow E_{n-1}^+|_{V_\lambda}$ (SRD-331 Lemma 3.2 / r=1 paper Lemma 2.4). For each SYT $T' \in \text{SYT}(\mu_1)$, $\Phi_*(v_{T'})$ is the normalised $+q$ -eigenvector of T_{n-1} on the $\{T_A, T_B\}$ 2-block, where T_A puts n at c_* , $n-1$ at c_1 , and T_B swaps. Φ_* is T_j -equivariant for $j \leq n-3$ (index gap ≥ 2 from T_{n-1}).

Recall: SRD-331 reduction formula.

Theorem 1 (SRD-331 Corollary 4.3). *For $\lambda = (3, 3, 1^q)$, $q \geq 2$,*

$$B_{\ell+1}^{(\lambda)} V_\lambda = (T_\ell+1)(T_{\ell+1}+1)(T_{n-2}+1) \Phi_* \left(B_{\ell(\mu_1)+1}^{(\mu_1)} V_{\mu_1} \right). \quad (1)$$

Equivalently, with the three outer indices $\ell = n-4$, $\ell+1 = n-3$, $n-2$:

$$B^{(\lambda)} = \mathcal{P} \Phi_*(B^{(\mu_1)}), \quad \mathcal{P} := (T_{n-4}+1)(T_{n-3}+1)(T_{n-2}+1).$$

Recall: the μ_1 -dim.

Theorem 2 (May-12 evening Theorem 5.4). *For $\mu_1 = (3, 2, 1^{q-1})$ with $|\mu_1| = q+4 \geq 8$ (i.e., $q \geq 4$),*

$$\dim B_{\ell(\mu_1)+1}^{(\mu_1)} V_{\mu_1} = 2.$$

Main result.

Theorem 3 (Main). *For $\lambda = (3, 3, 1^q)$ with $q \geq 4$,*

$$\dim B_{\ell+1}^{(\lambda)} V_\lambda = 2. \quad (2)$$

By Theorem ??, $B^{(\lambda)} = \mathcal{P} \Phi_*(B^{(\mu_1)})$. Hence

$$\dim B^{(\lambda)} \leq \dim \Phi_*(B^{(\mu_1)}) = \dim B^{(\mu_1)} = 2$$

(using Φ_* injective by Lemma ??/SRD-331 Lemma 2.4 and Theorem ??). This is the upper bound, already in SRD-331. The new content is the matching lower bound, which we prove in §?? via outer-factor injectivity.

2 The two ingredients: image landing and adjacent disjointness

2.1 Image of (T_j+1) lands in E_j^+

Lemma 4 (Image landing). *For every $H_q(S_n)$ -module V and every $1 \leq j \leq n-1$,*

$$(T_j+1)V \subseteq E_j^+(V).$$

Proof. The Hecke quadratic relation $(T_j - q)(T_j + 1) = 0$ gives $T_j(T_j + 1) = q(T_j + 1)$ in $H_q(S_n)$. Hence for any $x \in V$, $T_j(T_j + 1)x = q(T_j + 1)x$, i.e., $(T_j + 1)x \in E_j^+(V)$. $\square$

2.2 The May-19 adjacent-disjointness lemma

Lemma 5 (May-19 adjacent disjointness). *For every $H_q(S_n)$ -module V and every $1 \leq j \leq n-2$,*

$$E_j^+(V) \cap E_{j+1}^-(V) = 0, \quad E_j^-(V) \cap E_{j+1}^+(V) = 0.$$

Proof. This is 2026-05-19-phase-a-general.tex Lemma 3.3 (the $\dim E_j^+ \cap E_{j+1}^-$ direction), plus its symmetrically swapped form. Both are proved by restricting to the subalgebra $\langle T_j, T_{j+1} \rangle$, a quotient of $H_q(S_3)$, and checking the three irreducibles:

- **Trivial** (T_j, T_{j+1} both act as q): $E_j^- = E_{j+1}^- = 0$, so both intersections are trivially 0.
- **Sign** (both act as -1): $E_j^+ = E_{j+1}^+ = 0$, both intersections trivially 0.
- **Standard** (2-dim; in a basis where T_j is diagonal $\text{diag}(q, -1)$, T_{j+1} acts as a Hoefsmit 2-block at axial distance d_0 with $|d_0| \geq 2$, hence non-degenerate with non-zero off-diagonal entries). E_j^+ is spanned by the first basis vector v_a ; E_j^- by the second v_b . The eigenvectors of T_{j+1} are non-trivial linear combinations of v_a, v_b (non-degeneracy). So $v_a \notin E_{j+1}^-$ (else $T_{j+1}v_a = -v_a$, requiring the off-diagonal entry $b \mapsto a$ to vanish), and $v_b \notin E_{j+1}^+$ (similarly). Hence both intersections are 0.

The lemma applies to all V since any $\langle T_j, T_{j+1} \rangle$ -submodule decomposes into the three irreducibles above. $\square$

2.3 Synthesis: $(T_j + 1)$ is injective on E_{j+1}^+

Proposition 6 (Adjacent injectivity). *For every $H_q(S_n)$ -module V and every $1 \leq j \leq n-2$, the restricted map*

$$(T_j + 1)|_{E_{j+1}^+(V)} : E_{j+1}^+(V) \rightarrow E_j^+(V)$$

is well-defined (Lemma ??) and injective (Lemma ??).

Proof. The image lies in $E_j^+(V)$ by Lemma ??. The kernel of $T_j + 1$ on V is the (-1) -eigenspace $E_j^-(V)$ (since $T_j = -1$ on E_j^- makes $T_j + 1 = 0$ there, and on E_j^+ , $T_j + 1 = q + 1 \neq 0$). Hence the kernel of the restricted map is $E_j^-(V) \cap E_{j+1}^+(V)$, which is 0 by Lemma ??. $\square$

3 Outer-factor injectivity and the lower bound

Theorem 7 (Outer-chain injectivity). *For $\lambda = (3, 3, 1^q)$ with $q \geq 4$, the operator*

$$\mathcal{P} := (T_{n-4} + 1)(T_{n-3} + 1)(T_{n-2} + 1)$$

acts injectively on $E_{n-1}^+|_{V_\lambda}$, with image in $E_{n-4}^+|_{V_\lambda}$.

Proof. By iterated application of Proposition ??, the three restricted maps

$$(T_{n-2} + 1)|_{E_{n-1}^+} : E_{n-1}^+ \rightarrow E_{n-2}^+, \quad (T_{n-3} + 1)|_{E_{n-2}^+} : E_{n-2}^+ \rightarrow E_{n-3}^+, \quad (T_{n-4} + 1)|_{E_{n-3}^+} : E_{n-3}^+ \rightarrow E_{n-4}^+$$

are all well-defined and injective. The composition $(T_{n-4} + 1)(T_{n-3} + 1)(T_{n-2} + 1)|_{E_{n-1}^+}$ is therefore well-defined as a map $E_{n-1}^+ \rightarrow E_{n-4}^+$ and injective as a composition of injections. $\square$

Remark 8. A subtle point: the operator $\mathcal{P}$ is defined on all of V_λ , not just E_{n-1}^+ . The statement of Theorem ?? is about the RESTRICTION of $\mathcal{P}$ to E_{n-1}^+ . The proof works because each intermediate image $(T_{n-2} + 1) E_{n-1}^+ \subseteq E_{n-2}^+$ stays inside the domain of the next factor’s “adjacent injectivity” statement.

Proof of Theorem ??. The upper bound $\dim B^{(\lambda)} \leq 2$ is the SRD-331 result (§??). For the lower bound: by Theorem ??,

$$B^{(\lambda)} = \mathcal{P} \Phi_*(B^{(\mu_1)}).$$

$\Phi_*(B^{(\mu_1)})$ is a 2-dimensional subspace of $E_{n-1}^+|_{V_\lambda}$ (2-dim by Theorem ?? and Φ_* injectivity; contained in E_{n-1}^+ by construction of Φ_*). By Theorem ??, $\mathcal{P}$ acts injectively on this 2-dimensional subspace. Hence

$$\dim B^{(\lambda)} = \dim \mathcal{P}(\Phi_*(B^{(\mu_1)})) = \dim \Phi_*(B^{(\mu_1)}) = 2.$$

□

4 Generalisation to other $r=1$ shapes

The proof of Theorem ?? uses three shape-sensitive inputs:

1. The SRD-style reduction formula (Theorem ??) relating $B^{(\lambda)}$ to $\mathcal{P} \Phi_*(B^{(\mu_1)})$. This holds for any $r = 1$ rank-zero λ where the same-row-dies conjecture is proved on the same-row part of E_{n-1}^+ (SRD framework theorem). For $(3, 3, 1^q)$ this is SRD-331.
2. Non-degeneracy of the corner-pair $\{c_1, c_*\}$ 2-block ($|d| \geq 2$), which gives Φ_* .
3. The dim of $B^{(\mu_1)}$.

The injectivity argument (Proposition ?? and the chain Theorem ??) is *shape-agnostic*: it uses only the universal adjacent-disjointness lemma and the image-landing lemma, both of which hold in any $H_q(S_n)$ -module.

Theorem 9 (General $r=1$ dim equality). *Let $\lambda \vdash n$ be a rank-zero partition with exactly one length-preserving corner c_1 , column-1 bottom c_* , and $|c(c_*) - c(c_1)| \geq 2$. Assume the same-row-dies conjecture (SRD-331 Conjecture 5.1) holds for λ , so the reduction formula (??) applies with $\mathcal{P} = (T_\ell+1)(T_{\ell+1}+1) \cdots (T_{n-2}+1)$ ($n-1-\ell$ factors) and $\mu_1 = \lambda \setminus \{c_1, c_*\}$. Then*

$$\dim B_{\ell+1}^{(\lambda)} V_\lambda = \dim B_{\ell(\mu_1)+1}^{(\mu_1)} V_{\mu_1}.$$

Proof. The reduction formula gives $\dim B^{(\lambda)} \leq \dim B^{(\mu_1)}$ (via Φ_* injective and dimension-non-increase under $\mathcal{P}$). For the reverse: $\Phi_*(B^{(\mu_1)}) \subseteq E_{n-1}^+|_{V_\lambda}$, and the iterated $(T_j + 1)$ -chain $(T_\ell + 1) \cdots (T_{n-2} + 1)$ runs through indices $\ell, \ell + 1, \dots, n - 2$ in increasing order. Iteratively applying Proposition ??: $(T_{n-2} + 1) : E_{n-1}^+ \rightarrow E_{n-2}^+$ is injective; then $(T_{n-3} + 1) : E_{n-2}^+ \rightarrow E_{n-3}^+$ is injective; $\dots$; finally $(T_\ell + 1) : E_{\ell+1}^+ \rightarrow E_\ell^+$ is injective. Composition injective. Hence $\dim B^{(\lambda)} = \dim \mathcal{P} \Phi_*(B^{(\mu_1)}) = \dim B^{(\mu_1)}$. □

Remark 10 (Application range). Theorem ?? immediately gives $\dim B$ in closed form for any $r = 1$ shape where SRD-style same-row-dies has been established and the μ_1 -dim is known. Currently the SRD framework is established at:

- Hooks $(k, 1^{n-k})$, with $\mu_1 = (k-2, 1^{n-k})$. Inductive on k . (Closed by the Shift Lemma + hook j-formula chain.)
- $(3, 3, 1^q)$, with $\mu_1 = (3, 2, 1^{q-1})$. SRD-331.

- $(4, 2, 1^q)$, with $\mu_1 = (4, 1^q)$ via SRD-421 same-row analysis. The j-formula in this case is the hook $(4, 1^q)$ formula, $\dim = 1$ (hook dim-1, May-13).
- In general, $(k, 2, 1^q)$ for various k (recursive).

5 Computational verification

Mod- P verification at $P = 100\,003$, $q = 11$ (the seed specialisation), for $\lambda = (3, 3, 1^q)$ at $q \in \{4, 5, 6\}$:

- Adjacent-disjointness $E_j^- \cap E_{j+1}^+ = 0$ at $j \in \{n-4, n-3, n-2\}$: verified at each q .
- The outer-chain rank: $\text{rank}(\mathcal{P}|_{E_{n-1}^+}) = \dim E_{n-1}^+|_{V_\lambda}$ at each q . Specifically:

q	n	$\dim E_{n-1}^+ _{V_\lambda}$
4	10	85
5	11	133
6	12	196

- Direct dim: $\dim B^{(\lambda)} = 2$ at each q via direct computation of $R'_{\ell+1} \cdots R'_n V_\lambda$.

Scripts: `~/projects/scratch/2026-05-13-dim2-331-lower/`.

6 Honest status

What we proved. $\dim B^{(\lambda)} = 2$ structurally for $\lambda = (3, 3, 1^q)$ at every $q \geq 4$, closing the lower-bound gap of SRD-331 Remark 4.6.

What we generalised. An $r=1$ dim-equality $\dim B^{(\lambda)} = \dim B^{(\mu_1)}$ (Theorem ??) under the SRD-style hypothesis. This is shape-agnostic and uses only universal Hecke-algebra facts (image landing + adjacent disjointness).

What this clarifies. The earlier “stuck” position analysis in `2026-05-13-dim2-331/proof-sketch.md` (the natural attempt: position-of- $(n-2)$ separator preserved by outer factors; fails because T_{n-3}, T_{n-2} move letter $n-2$) was looking in the wrong direction. The correct argument doesn’t track positions at all — it uses the abstract $H_q(S_n)$ -module-structural fact that adjacent T_j -eigenspaces of opposite sign intersect trivially.

What this does NOT generalise to. Multi-corner ($r \geq 2$) shapes. There, the reduction formula has multiple Φ_X -branches, and the lower bound requires a separation argument between branches (see May-12 evening for $r = 2$ at $(3, 2, 1^*)$). The adjacent- injectivity argument shows that each individual $\Phi_X(B^{(\mu_X)})$ - branch is preserved by the outer chain, but the r branches may *collide* after the outer factors. So $\dim B^{(\lambda)} \leq r \cdot \dim B^{(\mu_X)}$, not equal in general. Separation between branches needs a different tool (position-of- n , May-12 evening template).

Status of related conjectures.

- For $(3, 3, 1^q)$, $q \in \{2, 3\}$: same-row-dies holds (SRD-331); $\dim B^{(\mu_1)} = 2$ at $q \in \{2, 3\}$ is computational (May-12 evening rigorous at $q \geq 4$). The current theorem is unconditional at $q \geq 4$; at $q \in \{2, 3\}$ it gives $\dim B^{(\lambda)} = 2$ *conditional on the computational μ_1 -dim*.

- For the $(4, 3, 1^q)$ family ($r = 2$, $\dim = 5$): the within-class lower-bound problem reduces to two tracer-pair computations on $\{F_A^1, F_A^2\}$ and $\{F_B^1, F_B^2, F_B^3\}$ separately (2026-05-13-4-3-1n-lower-bound-analysis). The F_A -pair within-class problem is exactly the $(3, 3, 1^{q-1})$ dim-2 problem (now closed by this paper at $q - 1 \geq 4$, i.e., $q \geq 5$). The F_B -triple within-class problem is the $(4, 2, 1^{q-1})$ dim-3 problem, closed by SRD-421 + May-12 evening-3. So the $(4, 3, 1^q)$ structural dim-5 lower bound reduces to two already-closed inner inputs, plus the outer A vs B decoupling already proved in the lower-bound-analysis paper. **Corollary (below).**

Corollary 11 (Within-class inputs for $(4, 3, 1^q)$). *Combined with the $(4, 3, 1^q)$ lower-bound-analysis paper, the results of this paper close one of the two within-class lower-bound inputs (specifically: $\dim B^{(\mu_A)} = 2$ for $\mu_A = (3, 3, 1^{q-1})$ at $q \geq 5$, i.e., for $\lambda = (4, 3, 1^q)$ at $q \geq 5$). The remaining input ($\dim B^{(\mu_B)} = 3$ for $\mu_B = (4, 2, 1^{q-1})$) is SRD-421 + May-12 evening-3, also closed.*

However, the $(4, 3, 1^q)$ lower-bound-analysis paper still requires the within-class lifting through the OUTER $\mathcal{O}_\lambda$ chain to be injective on the lifted inner basis vectors. The argument of the present paper (adjacent-injectivity) extends directly to this lifting — the outer chain $\mathcal{O}_\lambda$ at $\lambda = (4, 3, 1^q)$ is again a product of $(T_j + 1)$ ’s for j ranging up to $n - 2$, and the same chain of $E_j^+ \rightarrow E_{j-1}^+$ injectivity arguments applies.

Hence by combining (i) the $(4, 3, 1^q)$ -LB-analysis-AvsBdecoupling, (ii) the inner-basis dimensions from SRD-331-corollary and SRD-421+evening-3, and (iii) the adjacent-injectivity propagation of this paper, we get the structural 5 at every q such that all three pieces are unconditional.

Combined with the May-13 morning upper bound,

$$\dim B^{((4,3,1^q))} = 5$$

structurally at $q \geq q_$ for some explicit threshold (the maximum of the underlying structural thresholds: $q \geq 5$ for SRD-331’s μ_A piece, $q \geq ?$ for SRD-421’s μ_B piece, etc.).*

Remark 12. Corollary ?? is stated conditionally on the literal chaining of arguments through the outer factors $\mathcal{O}_\lambda$ of $\lambda = (4, 3, 1^q)$. The full unconditional statement (“ $\dim B = 5$ at all $q \geq q_*$ for explicit q_* ”) requires a careful audit of which inputs apply at which q . The present paper does not perform that audit; it suggests the adjacent-injectivity propagation as the unifying conceptual ingredient.

7 What’s next

Immediate corollaries (not in this paper). The adjacent-injectivity framework gives an immediate dim-equality $\dim B^{(\lambda)} = \dim B^{(\mu_1)}$ for any $r=1$ shape with SRD-style same-row-dies. This subsumes:

- Hooks $(k, 1^{n-k})$ at $k \geq 3$ ($\mu_1 = (k - 2, 1^{n-k})$): $\dim B^{(\lambda)} = \dim B^{(\mu_1)}$. For hooks, $\dim B^{(\mu_1)}$ is given by hook-dim-1 paper (May-13). Recursive descent gives $\dim B^{(\lambda)} = 1$ for all hooks.
- $(k, 2, 1^q)$ via SRD-421 framework: same-row-dies on row 1, $\mu_1 = (k, 1^{q+1})$. Reduces to hook μ_1 .
- $(k, k, 1^q)$ at $k = 3$ (this paper).

Deeper question. The adjacent-injectivity argument suggests a structural fact: the operator $\mathcal{P} = (T_{n-a} + 1) \cdots (T_{n-2} + 1)$ (for any number of factors) is *injective* on $E_{n-1}^+|_V$ for any $H_q(S_n)$ -module V . This is a clean general fact independent of the rank-zero programme; it says that

the iterated $+q$ -eigenspace descent through adjacent indices loses no dimension. The proof is the straightforward chain of adjacent-disjointness arguments.

This in turn suggests that for any reduction formula of the SRD-style, $\dim B^{(\lambda)}$ is determined entirely by $\dim B^{(\mu_1)}$ plus the dim of the same-row-dies part (if non-zero). For $r = 1$ shapes with the same-row part dying, this gives the cleanest possible reduction.